Fabrizio Sisinni

Curriculum Vitae

CRYPTO Lab
AI4I
Torino, Italy
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0009-0007-9641-4329



I am a mathematician specialized in lattice-based cryptography, and provable security. My research focuses on the provable security of lattice-based schemes in both the Random Oracle and Quantum Random Oracle Models. I am particularly interested in the study of hardness assumptions over rings and modules, such as the Ring Learning With Errors and Module Learning With Errors problems.

Professional Experience

01/11/2026– **Postdoc**, *The Italian Institute of Artificial Intelligence*, Torino
CRYPTO Lab

01/12/2025–31/10/2026 **Postdoc**, *Technical University of Denmark*, Copenhagen
Post-Quantum Key Agreement

Education

01/12/2022–30/11/2025 **PhD Student (Awarded 24/03/2026)**, *Technical University of Denmark*, Copenhagen
Marie Skłodowska-Curie Doctoral Network – Quantum-Safe Internet

29/10/2021 **Master's Degree in Mathematics**, *University of Pisa*, Pisa, *110/110 cum laude*
Curriculum in Algebra and Cryptography

26/10/2018 **Bachelor's Degree in Mathematics**, *University of Pisa*, Pisa, *106/110*
Theoretical Curriculum

Short-term Research Visits

06/2025–07/2025 **Secondment**, *Joppe W. Bos*, NXP Semiconductors, Leuven, Belgium

01/2025–02/2025 **Secondment**, *Christian Schaffner*, Centrum Wiskunde & Informatica, Amsterdam, Netherlands

01/2024–03/2024 **Secondment**, *Andreas Hülsing & Kathrin Hövelmanns*, Eindhoven University of Technology, Eindhoven, Netherlands

Relevant Educational Experience

2025 **Reading group on Quantum Computing**, *Technical University of Denmark*

2024 **IACR Summer school in Warsaw**, *IDEAS NCBR, University of Warsaw*

2024 **QSI Summer school on PQC**, *Eindhoven University of Technology*

2024 **QSI Winter school on QKD**, *University of Padova*

Student Supervision

2025 **Hanna Takeuchi Thirstrup**, *Post Quantum Cryptography and Decryption Failures*

Talks

2026 **SAC 2026**, *IACR*, Ottawa
Provable decryption failure security for practical lattice-based PKE

2026 **ICQCS 2026**, *QSI, QuanTiP*, Paris
Provable decryption failure security for practical lattice-based PKE

2026 **Nordicrypt Spring 2026**, *DTU*, Copenhagen
Provable decryption failure security for practical lattice-based PKE

2025 **EUROCRYPT 2025**, *IACR*, Madrid
(Un)-breakable curses - re-encryption in the Fujisaki-Okamoto transformation

2025 **Nordicrypt Spring 2025**, *ITU*, Copenhagen
(Un)-breakable curses - re-encryption in the Fujisaki-Okamoto transformation

2024 **CRYPTO 2024**, *IACR*, Santa Barbara
Provable security against decryption failure attacks from LWE

2024 **Nordicrypt Spring 2024**, *DTU*, Kongens Lyngby
Provable security against decryption failure attacks from LWE

Teaching

2023/2024 **Teaching Assistant**, *DTU*, Applied Cryptography

2023/2024 **Teaching Assistant**, *DTU*, Cryptology 1

2024 **Trainer for the Danish Cybersecurity Challenge**, *DTU*, Cryptology training

Community Service

2027 **Program Committee**, *CT-RSA 2027*

2026 **Session Chair**, *QSI, QuanTiP*, ICQCS 2026

2025 **(Co-)Organizer**, *DTU*, QSI Workshop in Copenhagen

2025 **Reviewer**, *DTU*, QSI Workshop in Copenhagen

2023 **Sub Reviewer**, *ACM Conference on Computer and Communications Security (CCS)*

2024–2026 **Sub Reviewer**, *TCC, PKC, EUROCRYPT, ASIACRYPT*

Publications

Peer-reviewed Publications

– Hövelmanns, K., Hülsing, A., Majenz, C., **Fabrizio Sisinni**: (un)breakable curses - re-encryption in the fujisaki-okamoto transform. In: Advances in Cryptology - EUROCRYPT 2025. Lecture Notes in Computer Science, vol. 15602. Springer, `https://doi.org/10.1007/978-3-031-91124-8_9`

– Majenz, C., **Fabrizio Sisinni**: Provable decryption failure security for practical lattice-based PKE. IACR Cryptol. ePrint Arch. `https://eprint.iacr.org/2025/1997`

– Majenz, C., **Fabrizio Sisinni**: Provable security against decryption failure attacks

from LWE. In: Advances in Cryptology - CRYPTO 2024. Lecture Notes in Computer Science, vol. 14921. Springer, https://doi.org/10.1007/978-3-031-68379-4_14